Financial facts need their full context

Independent research program 31 October 2026 UTC

Not peer reviewed. One current SEC extract; fixed relational-key comparisons.

Practical finding

A company, tag and period-end date do not uniquely identify a financial fact. In this fixed cohort, that short key collapses 247 facts into 27 groups. A one-row-per-key join would discard 220 distinct source contexts.

The short key mixes durations in 10 groups and segment/coregistrant contexts in 16 groups. Adding duration still leaves 210 extra contexts within 37 groups. Retaining segment/coregistrant identity separates every observed fact here. Accession remains part of the full source key even though each company has only one selected filing in this archive.

An authored naive-winner simulation finds 215 source facts with a different numeric value from their short-key winner. Different durations or dimensions can describe different quantities. This is not a count of verified restatements, financial losses or historical feature leakage.

Recommended use

Carry accession, tag, taxonomy version, period, duration, unit, segment and coregistrant identity through a financial-data join. Require a complete context before returning a qualified fact; abstain when it is absent or ambiguous. Classify context differences before interpreting a changed value as a revision.

The implementation exercise retains a useful failed assumption: an Apple example requested a 2026 taxonomy, while the source uses a 2025 taxonomy. The lookup correctly abstains. A separately labeled post-analysis positive control uses the actual version and returns a qualified source value. Filing year is insufficient evidence for taxonomy version.

Fixed source and scope

The official current 2026 Q3 face-financial archive supplies 3 selected filings from Apple, Microsoft and Walmart. The protocol fixed those CIKs, six standard financial tags and all six key specifications before retrieval. All 18 company/tag cells are retained, including empty cells; this is a convenience cohort.

The original broad historical restatement/model-leakage question remains unexecuted. There are 0 repeated-context observations across accessions in this selected cohort. That absence cannot establish that historical financial values are stable.

Complete fixed key accounts

Key	Groups	Extra contexts	Conflicts	Value differences	Mixed duration	Mixed segment
K0	27	220	15	215	10	16
K1	37	210	17	203	0	24
K2	37	210	17	203	0	24
K3	37	210	17	203	0	24
K4	247	0	0	0	0	0
K5	247	0	0	0	0	0

K0 uses CIK, tag and rounded end date. K1 adds duration, K2 unit, K3 taxonomy, K4 segment/coregistrant identity and K5 accession. All units and taxonomy versions remain explicit even where they do not split a group in this cohort. Conflicting-group counts need not decline at every step: a larger group can split into several groups that each contain differing values.

All company/tag cells, including zero coverage

Company	Tag label	Facts	K0 groups	K0 extra	K0 value differences	K4 groups
Walmart	Assets	11	3	8	8	11
Walmart	Liabilities	0	0	0	0	0
Walmart	Net income	4	2	2	2	4
Walmart	Contract revenue	88	2	86	86	88
Walmart	Revenues	16	2	14	14	16
Walmart	Net sales	0	0	0	0	0
Apple	Assets	2	2	0	0	2
Apple	Liabilities	2	2	0	0	2
Apple	Net income	8	2	6	4	8
Apple	Contract revenue	48	2	46	46	48
Apple	Revenues	0	0	0	0	0
Apple	Net sales	0	0	0	0	0
Microsoft	Assets	2	2	0	0	2
Microsoft	Liabilities	2	2	0	0	2
Microsoft	Net income	6	3	3	0	6
Microsoft	Contract revenue	58	3	55	55	58
Microsoft	Revenues	0	0	0	0	0
Microsoft	Net sales	0	0	0	0	0

Tag labels are abbreviated for readability; the CSV retains the exact XBRL identifiers. Counts of zero indicate no selected facts for that company/tag, not a zero financial amount. The repository retains all 108 company/tag/key cells, 632 full group accounts and 1,482 memberships.

The naive winner is the lexicographic maximum of accepted timestamp, accession and complete context. This deterministic rule is authored for the audit, not a defensible financial selection policy. No dollar differences are pooled across incompatible quantities, and no source observation is silently averaged or removed.

Verification, implementation and limits

Independent evidence

The native archive controls cover 6,516 SUB rows, 87,871 TAG rows, 731,170 PRE rows and 3,905,397 NUM rows. Independent pandas C chunks, SQLite key multiplicities, Fraction values and date parsing reconcile every selected fact in 373 finite conditions. Saved-source checks pass 2,232; complete independent relational/numeric checks pass 2,039.

The original committed freeze bound 24 actual dependencies before comparisons. One disclosed post-comparison checker correction replaces an incorrect expectation that every complete request must succeed. The original seven requests, production logic, source and all six primary output bytes remain unchanged. HISTORY preserves the original freeze, checker and output hashes; the additional positive control is explicitly post-analysis.

All 32 authored fixture methods pass. Separate full-history CPython 3.10.11 and 3.12.14 replays reproduce six primary files exactly, preserving all eight source-directory files. Both also reproduce and independently check the additional positive control. Current figure checks bind 54 bars and 216 actual SVG coordinate conditions. Finite checks are not universal correctness guarantees.

Runnable implementation exercise

The original seven requests return two abstentions and five rejections. One additional positive control returns a qualified current fact and passes 25 independent source/history checks. The interface retains units, duration, accession and explicit unknown source truth, original-time availability and formal-restatement status. It does not certify arbitrary caller input or a production service.

```
python -m pip install -r requirements.txt
python src/replay.py --out ../financial-context-replay
python src/positive_control.py --out ../positive-control
python src/check_positive_control.py
```

Use new output directories. Saved-data replay does not redownload mutable native sources.

Material limits

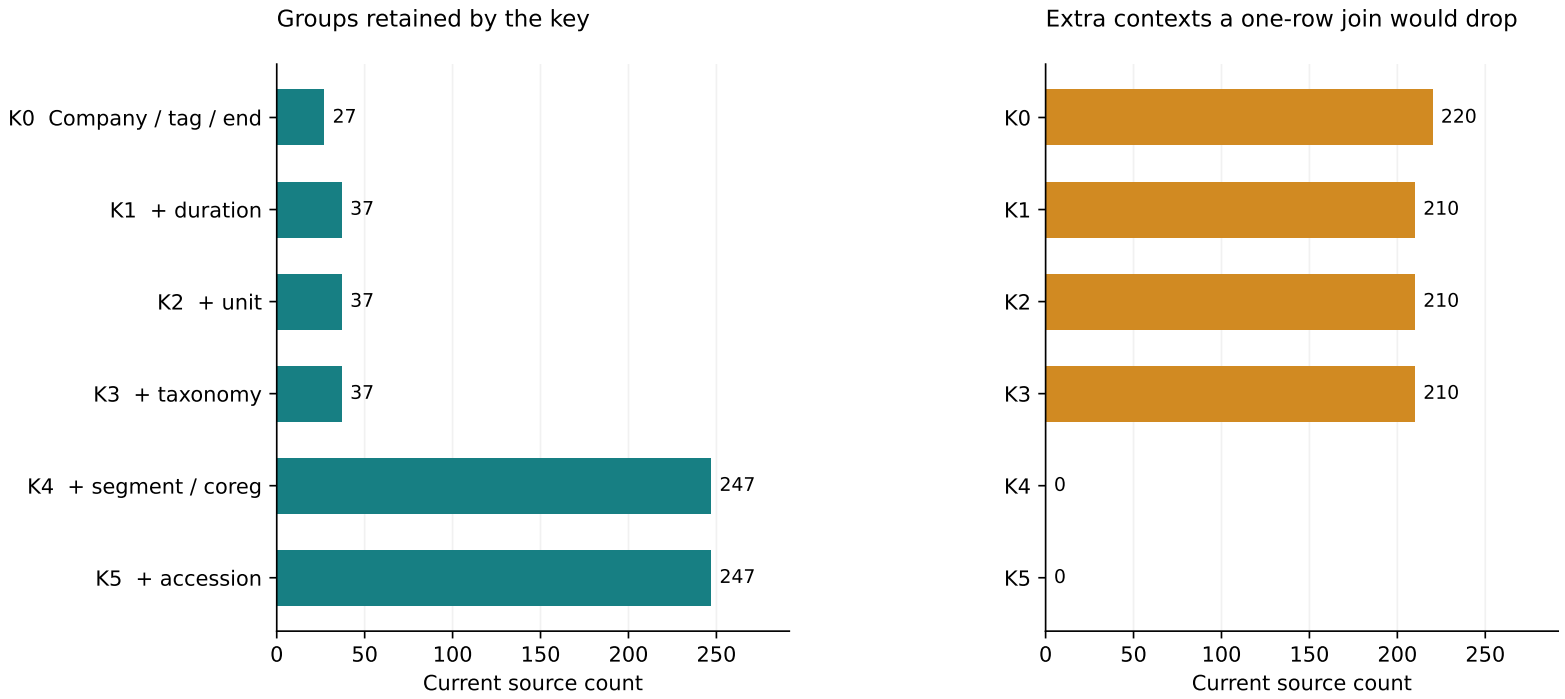
One quarter, three familiar filers and six tags do not represent all financial reporting. Source values were visible during reconciliation; this was source informed, not blinded. SEC reprocessed the current files with segment facts. Rounded month-end and quarter-duration fields do not recover exact fiscal boundaries. Accepted timestamps receive no verified timezone conversion or original-publication interpretation.

There is no verified formal restatement, predictive accuracy, investment advice, source-truth certification, new estimator, deployment or realized saving. Native archives, corporate report prose, footnotes, labels, business contact fields and EINs remain private. Separate formal-publication projects, including Azure optimization, remain private.

Source: SEC Financial Statement Data Sets, current 2026 Q3 archive, captured 31 October 2026 UTC. https://www.sec.gov/data-research/sec-markets-data/financial-statement-data-sets. Exact source/member hashes, all native table counts and earlier access/parser failures are retained. SEC warns that extracts may contain errors and do not replace full filings. SEC does not endorse this independent analysis. MIT covers original code; minimal government/registrant facts are attributed separately.

A short financial key drops distinct source contexts

247 source facts stay fixed; all six declared key specifications are retained

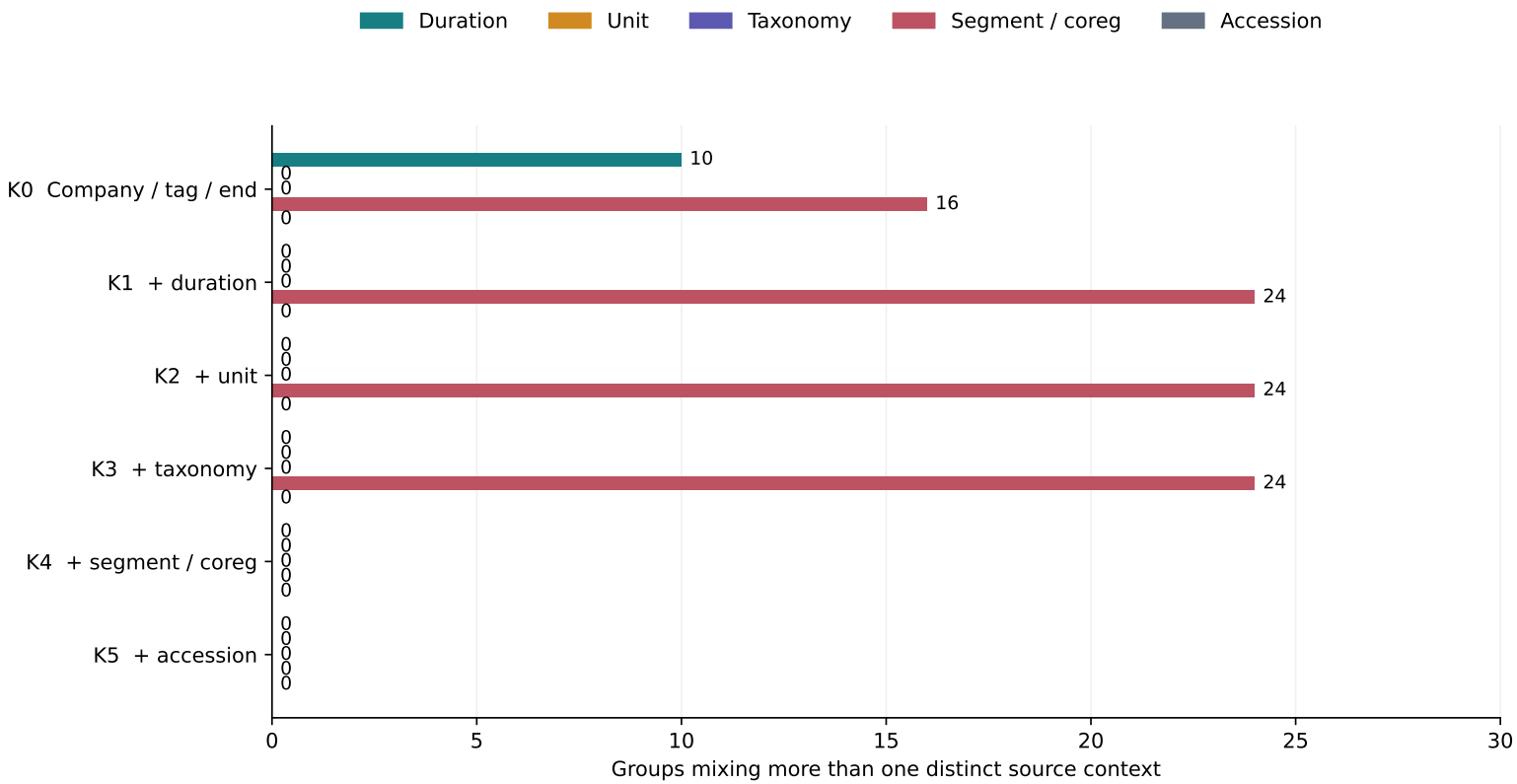


K4 matches the full key in this cohort because each company has one selected accession; accession remains part of source identity.

All omitted-context accounts

Duration and segment context explain the observed collisions

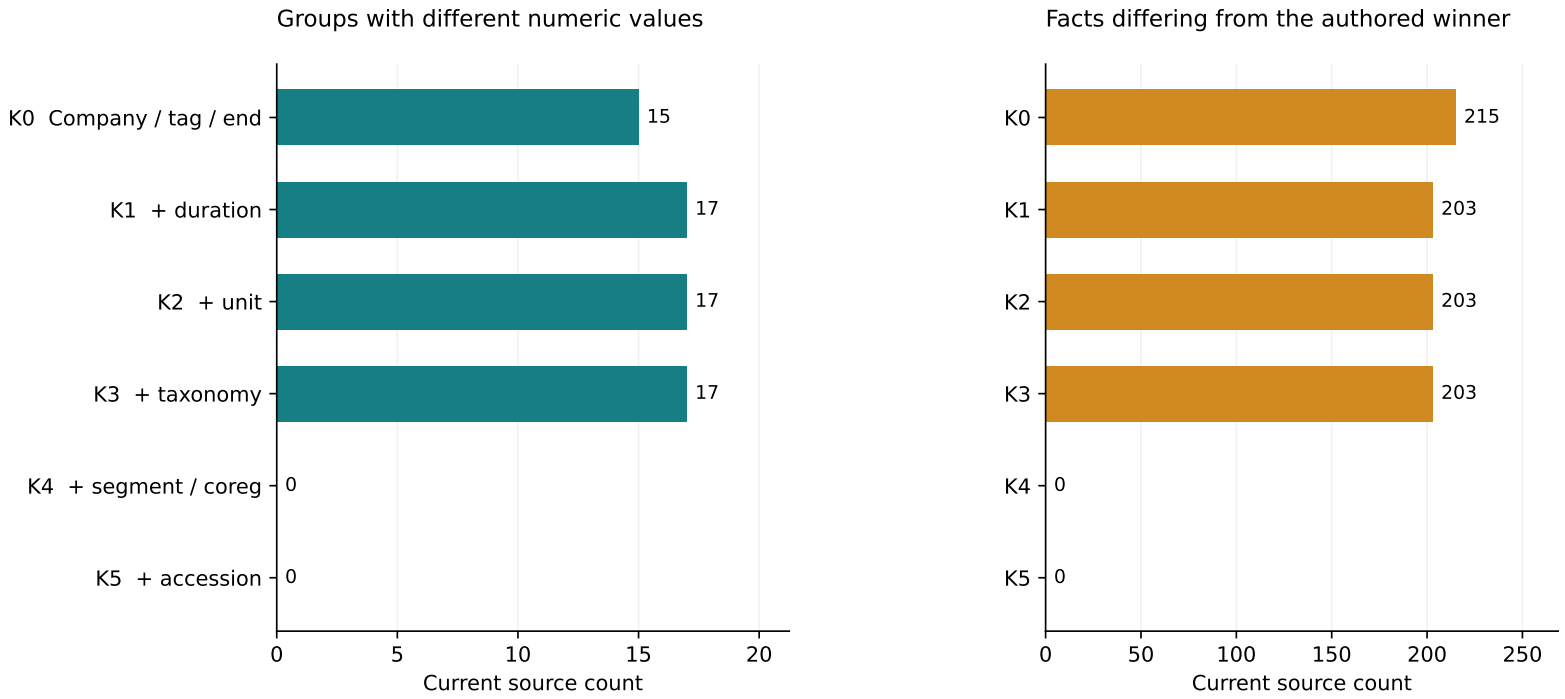
Counts refer to groups mixing each field; all zero categories remain visible



No unit, taxonomy or accession mixtures occur within the fixed base groups here; the source still requires those fields.

Different numeric values can describe different quantities

Authored naive-winner simulation; different durations or dimensions do not establish a formal restatement



Conflicting-group count need not fall at every step: adding duration can split one mixed group into several conflicting groups.